Web of dualities on non-orientable surfaces

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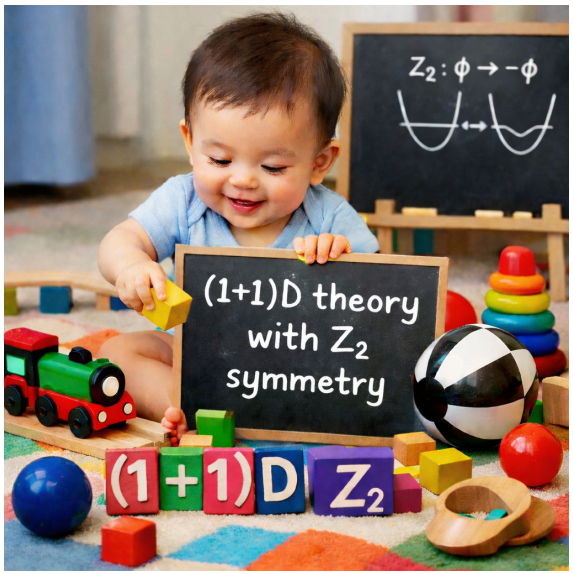
Based on [IO-Tsuji 2601.09067].

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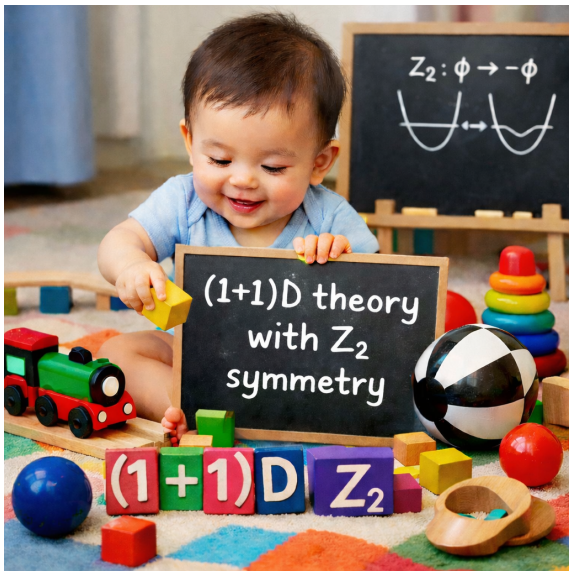


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- Gauging,
- Fermionizing,
- Stacking.

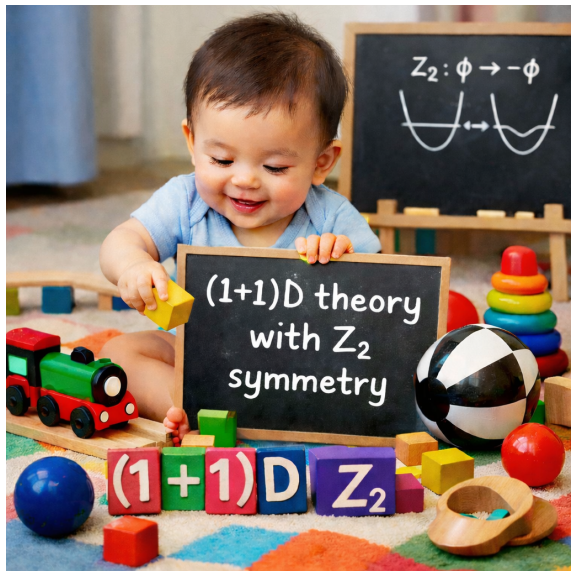


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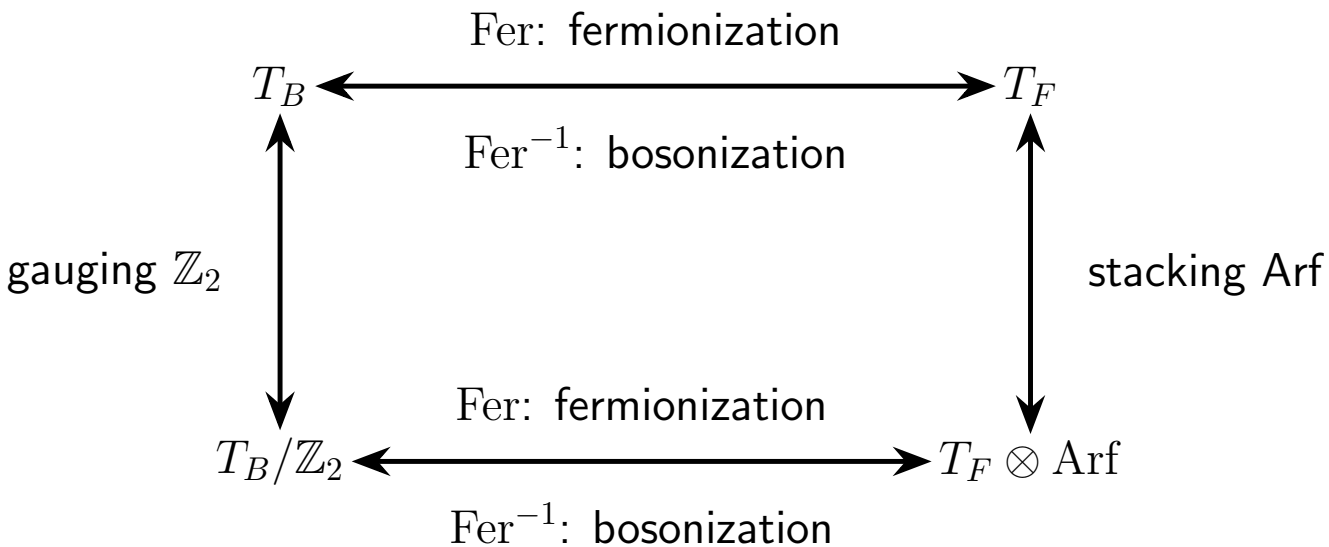
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$\rightarrow$ Duality web [Karch et al. 1902.05550]:



In general, it is understood that

QFT with reflection/time-reversal symmetry
$=$ QFT on non-orientable manifolds

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$\rightarrow$ Can we construct a non-orientable duality web?

$\rightarrow$ Yes! [Spieler 2504.20021]

$\rightarrow$ We study more about this duality web from several aspects:

- Symmetry TFT (SymTFT)
- Sectors of Hilbert space
- Group structure of topological manipulations

- Duality web on orientable surfaces
  - $\mathbb{Z}_2$ symmetry and its gauging
  - $\mathbb{Z}_2$-quadratic refinement, Arf invariant, and spin structures
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These are familiar concepts in QFT $\rightarrow$ What changes when $G = \mathbb{Z}_2$?

Fact:
For $G = \mathbb{Z}_2$, partition functions and a path integral are given as

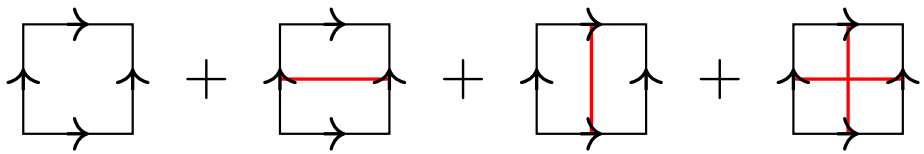
$$Z_{T_B}[a] \quad a \in H^1(\Sigma; \mathbb{Z}_2), \qquad \int \mathcal{D}A = \frac{1}{|H^1(\Sigma; \mathbb{Z}_2)|^{1/2}} \sum_{a \in H^1(\Sigma; \mathbb{Z}_2)}$$

Therefore, $\mathbb{Z}_2$ gauging is given as

$$Z_{T_B/\mathbb{Z}_2} = \frac{1}{|H^1(\Sigma; \mathbb{Z}_2)|^{1/2}} \sum_{a \in H^1(\Sigma; \mathbb{Z}_2)} Z_{T_B}[a]$$

e.g. For $\Sigma = T^2$, with $H^1(T^2; \mathbb{Z}_2) \cong \{0, x, y, x+y\}$, we obtain

$$Z_{T_B/\mathbb{Z}_2} = \frac{1}{2}(Z_{T_B}[0] + Z_{T_B}[x] + Z_{T_B}[y] + Z_{T_B}[x+y])$$

$$= \frac{1}{2}($$  $$).$$

We can consider refined version of gauging so that resulting theory $T_B/\mathbb{Z}_2$ has dual $\mathbb{Z}_2$ symmetry:

$$Z_{O^B(T_B)}[a] := \frac{1}{|H^1(\Sigma;\mathbb{Z}_2)|^{1/2}} \sum_{b\in H^1(\Sigma;\mathbb{Z}_2)} (-1)^{\int a\cup b} Z_{T_B}[b]$$

Here, $a \in H^1(\Sigma;\mathbb{Z}_2)$ is a background field of dual $\mathbb{Z}_2$ symmetry and cup product $\cup$ is a discrete version of wedge product $\wedge$.

Remark:
$(O^B)^2 = 1$, i.e. gauging twice takes you back to the original theory.

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Fermionic theory depends on spin structures of the spacetime.

e.g. We have four choices of spin structures on torus T^2;

$(x, y) = (\mathsf{NS}, \mathsf{NS}), (\mathsf{NS}, \mathsf{R}), (\mathsf{R}, \mathsf{NS}), (\mathsf{R}, \mathsf{R})$.

Remark: Orientable genus g surface Σ_g admits 2^{2g} spin structures.

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Fact:
$\{\text{Spin structure } \eta \text{ on } \Sigma\} \overset{1:1}{\longleftrightarrow} \{q_\eta : H^1(\Sigma; \mathbb{Z}_2) \to \mathbb{Z}_2 \text{ s.t. } \cdots\}$

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q_η is a $\mathbb{Z}_2$ valued function called a quadratic refinement.
e.g. Four choices of quadratic refinement on T^2;

$$(\mathsf{NS}, \mathsf{NS}) : q_\eta(0) = 0,\ q_\eta(x) = 0,\ q_\eta(y) = 0,\ q_\eta(x+y) = 1,$$
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Fact:
Spin structures are classified into two classes.

How do we classify these spin structures on T^2?

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Natural classification is $\{(\mathsf{NS},\mathsf{NS}),(\mathsf{NS},\mathsf{R}),(\mathsf{R},\mathsf{NS})\}$ and $\{(\mathsf{R},\mathsf{R})\}$.

Remark:
This classification is closed under $PSL(2,\mathbb{Z})$ i.e., $\tau \mapsto -1/\tau$, $\tau \mapsto \tau+1$.

Is there any way to classify them in general? $\rightarrow$ Arf invariant

Arf invariant

$$Z_{\mathrm{Arf}}[\eta] := \exp\left(\pi i \mathrm{Arf}(\eta)\right) = \frac{1}{|H^1(\Sigma;\mathbb{Z}_2)|^{1/2}} \sum_{a\in H^1(\Sigma;\mathbb{Z}_2)} \exp\left(\pi i q_\eta(a)\right)$$

e.g. Four choices of spin structures on T^2;

$$Z_{\mathrm{Arf}}[(\mathsf{NS},\mathsf{NS})] = \frac{1}{4^{1/2}}(1+1+1-1) = 1$$

$$Z_{\mathrm{Arf}}[(\mathsf{NS},\mathsf{R})] = \frac{1}{4^{1/2}}(1+1-1+1) = 1$$

$$Z_{\mathrm{Arf}}[(\mathsf{R},\mathsf{NS})] = \frac{1}{4^{1/2}}(1-1+1+1) = 1$$

$$Z_{\mathrm{Arf}}[(\mathsf{R},\mathsf{R})] = \frac{1}{4^{1/2}}(1-1-1-1) = -1$$

$\to$ Arf invariant classifies $\{(\mathsf{NS},\mathsf{NS}),(\mathsf{NS},\mathsf{R}),(\mathsf{R},\mathsf{NS})\}$ and $\{(\mathsf{R},\mathsf{R})\}$.

Physically Arf invariant is a partition function of the Kitaev chain.

Mathematically Arf invariant gives a homomorphism $\Omega_2^{\mathrm{Spin}}(\mathrm{pt}) \xrightarrow{\sim} \mathbb{Z}_2$.

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How do we construct a fermionic theory, i.e. a theory which depends on a spin structure η.

What we have now are

- Bosonic theory with $\mathbb{Z}_2$ symmetry $Z_{T_B}[a]$ where $a \in H^1(\Sigma; \mathbb{Z}_2)$
- Quadratic refinement $q_\eta : H^1(\Sigma; \mathbb{Z}_2) \to \mathbb{Z}_2$ which has a one-to-one correspondence with a spin structure η

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$\to$ Fermonization Fer : {bosonic theory} $\to$ {fermionic theory} is given as

$$Z_{\mathrm{Fer}(T_B)}[\eta] := \frac{1}{|H^1(\Sigma; \mathbb{Z}_2)|^{1/2}} \sum_{a \in H^1(\Sigma; \mathbb{Z}_2)} \exp(\pi i q_\eta(a)) Z_{T_B}[a]$$

$\to$ Bosonization Fer^{-1} : {fermionic theory} $\to$ {bosonic theory} is given as

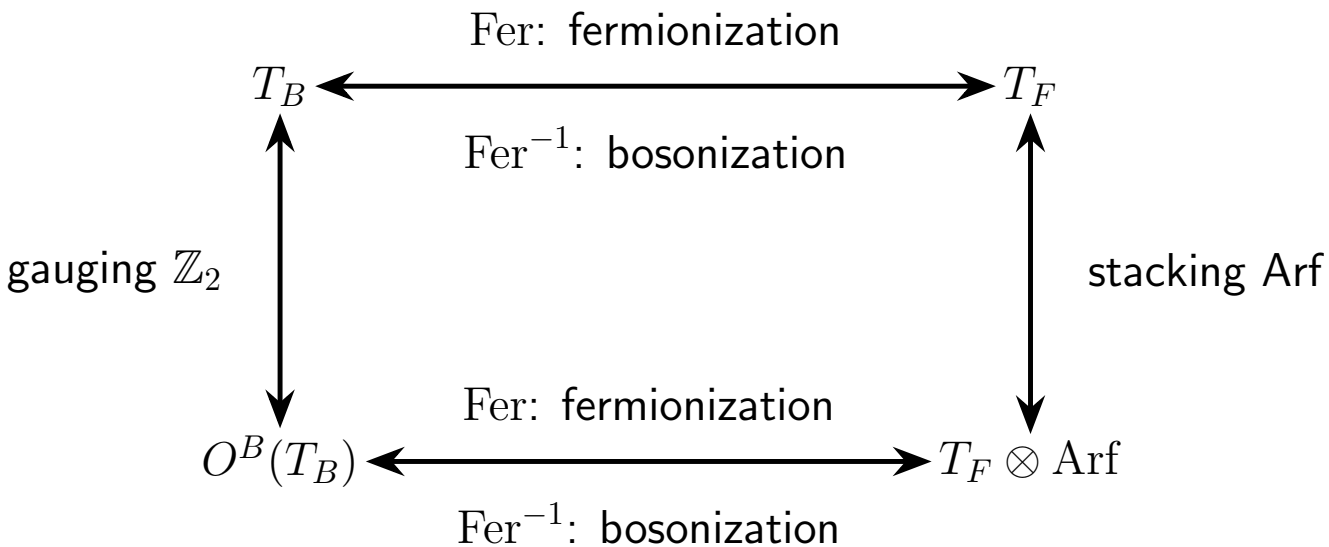
$$Z_{\mathrm{Fer}^{-1}(T_F)}[a] := \frac{1}{|H^1(\Sigma; \mathbb{Z}_2)|^{1/2}} \sum_{\eta \in \mathrm{Spin}(\Sigma)} \exp(\pi i q_\eta(a)) Z_{T_F}[\eta]$$

What corresponds to gauging O^B in fermionic theories?

i.e. $O^F := \mathrm{Fer} \circ O^B \circ \mathrm{Fer}^{-1} : \{\text{fermionic theory}\} \to \{\text{fermionic theory}\}$

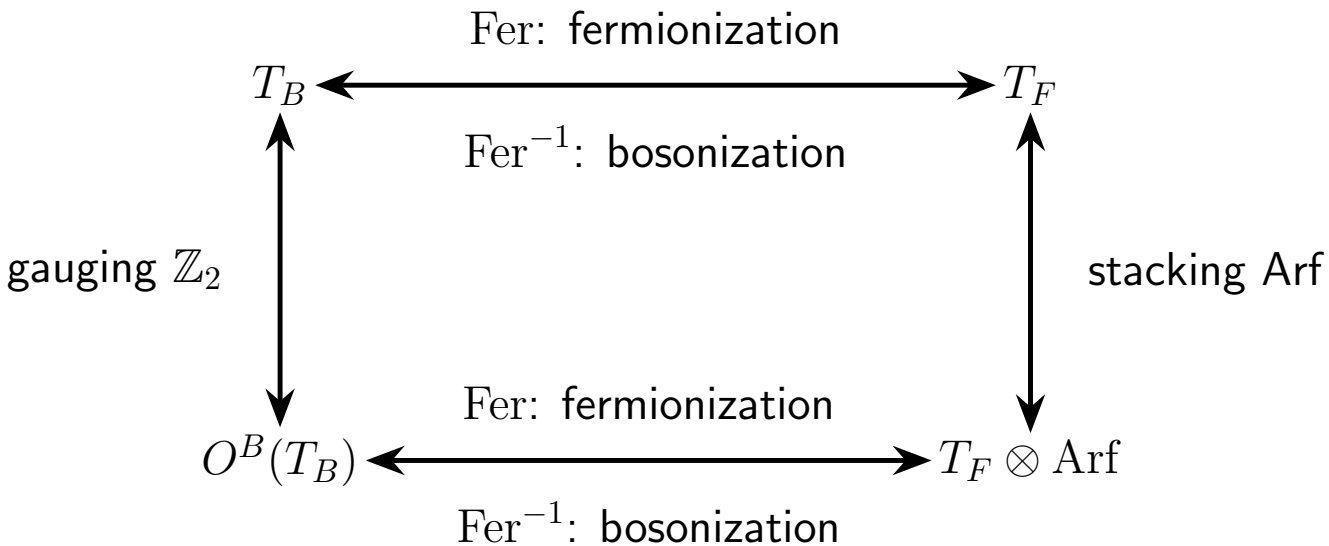
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One can check that

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Remark: $Z_{\mathrm{Arf}}[\eta] \in \mathbb{Z}_2 \Leftrightarrow (O^B)^2 = 1$

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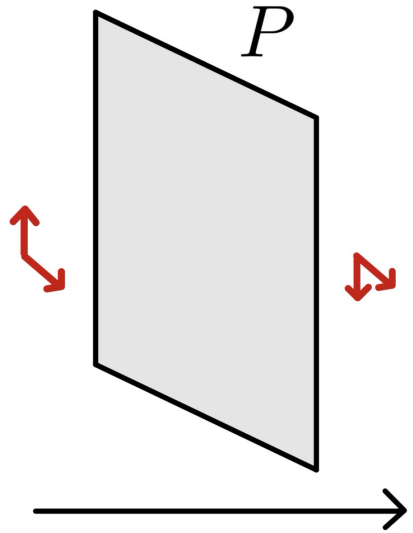
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$\rightarrow$ Existence of the orientation reversing defect $P \in H_d(M^{d+1}; \mathbb{Z}_2)$.

Using Poincaré duality, we can define [Thorngren 1404.4385]

$$w_1 := \mathrm{PD}(P) \in H^1(M^{d+1}; \mathbb{Z}_2).$$

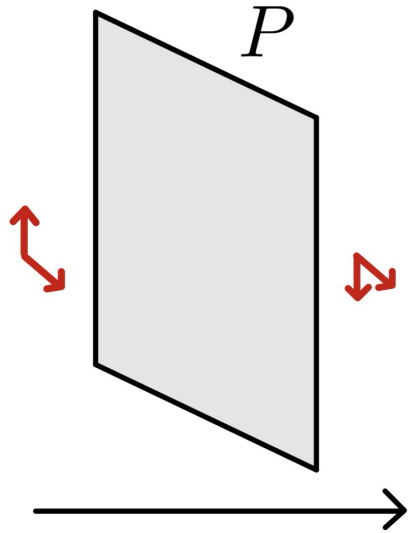


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Recall:

In $(3+1)$D Maxwell theory, we have

$$\int_\Sigma \frac{F}{2\pi} \in \mathbb{Z}.$$

If $\int_\Sigma \frac{F}{2\pi} \neq 0$, we cannot define $U(1)$ gauge field globally.

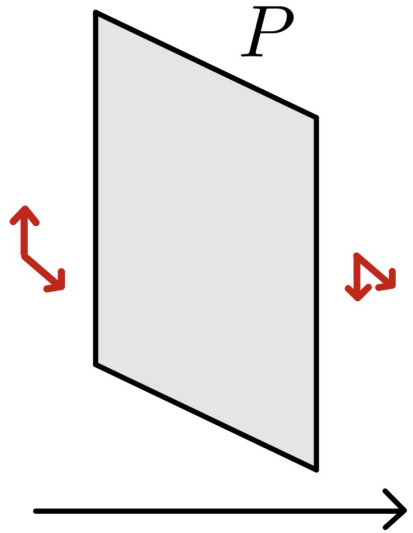
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Remark: w_1 is the obstruction to orientability.

In the presence of nontrivial w_1, a fermionic theory depends on Pin structure, i.e., non-orientable analogue of a spin structure.

Recall:

On an orientable surface, we take $R(\theta) \in SO(2)$, and let $\widetilde{R}(\theta) \in \mathrm{Spin}(2)$ be its lift. Then

$$R(2\pi) = 1, \qquad \widetilde{R}(2\pi) = (-1)^F.$$

Thus

$$\widetilde{R}(2\pi)\psi = \begin{cases} \psi & \text{if } \psi \text{ is bosonic,} \\ -\psi & \text{if } \psi \text{ is fermionic.} \end{cases}$$

On a non-orientable surface, reflections $P \in O(2)$ are allowed $P^2 = 1$.
However a lift $\widetilde{P}$ can satisfy either

$$\widetilde{P}^2 = 1 \text{ or } \widetilde{P}^2 = (-1)^F.$$

These two possibilities define the two double covers Pin^+ and Pin^-.
In this talk we use Pin^-, so the lifted reflection satisfies

$$\widetilde{P}^2 = (-1)^F.$$

Equivalently,

$$\widetilde{P}^2\psi = \begin{cases} \psi & \text{if } \psi \text{ is bosonic,} \\ -\psi & \text{if } \psi \text{ is fermionic.} \end{cases}$$

Hereafter, we write $\widetilde{P} \in \mathrm{Pin}^-(2)$ simply as P.

Fact:

- $\{\mathrm{Pin}^-$ structure η on $\Sigma\} \overset{1:1}{\longleftrightarrow} \{q_\eta : H^1(\Sigma; \mathbb{Z}_2) \to \mathbb{Z}_4$ s.t. $\cdots\}$
- Pin^- structures are classified into eight classes.
- ABK invariant valued in $\mathbb{Z}_8$ classifies them.

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$\to$ Stacking with SPT phases! $(-1)^{\int a\cup w_1}$, $(-1)^{\int w_1\cup w_1}$.

- SPT (Symmetry-Protected Topological) phase: a TQFT whose Hilbert space on any closed spatial manifold is one-dimensional.
- Stacking: At the level of partition functions, stacking corresponds to multiplication.

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$\to$ We can define operations S_1^B, S_2^B as follows:

$$Z_{S_1^B(T_B)}[a] := (-1)^{\int a\cup w_1} Z_{T_B}[a]$$

$$Z_{S_2^B(T_B)}[a] := (-1)^{\int w_1\cup w_1} Z_{T_B}[a]$$

	orientable	non-orientable
w_1	trivial	nontrivial
str	Spin	Pin^-
quad ref	$q_\eta : H^1(\Sigma;\mathbb{Z}_2) \to \mathbb{Z}_2$	$q_\eta : H^1(\Sigma;\mathbb{Z}_2) \to \mathbb{Z}_4$
Inv	$\mathrm{Arf}(\eta) \propto \sum_a e^{\pi i q_\eta(a)} \in \mathbb{Z}_2$	$\mathrm{ABK}(\eta) \propto \sum_a e^{\frac{\pi i}{2} q_\eta(a)} \in \mathbb{Z}_8$
Fer	$Z_{\mathrm{Fer}(T_B)} \propto \sum_a e^{\pi i q_\eta(a)} Z_{T_B}[a]$	$Z_{\mathrm{Fer}(T_B)} \propto \sum_a e^{\frac{\pi i}{2} q_\eta(a)} Z_{T_B}[a]$
SPT	trivial	$(-1)^{\int a \cup w_1}$, $(-1)^{\int w_1 \cup w_1}$

Table: Summary of orientable vs non-orientable

- Duality web on orientable surfaces
  - $\mathbb{Z}_2$ symmetry and its gauging
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Recall:

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$$\mathcal{F}(\psi)(p) := \tilde{\psi}(p) = \frac{1}{\sqrt{2\pi}} \int dx\, e^{-ipx} \psi(x).$$

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Remark:

$\mathcal{F}$ is a $\mathbb{Z}_2$ operation $\mathcal{F} \circ \mathcal{F}(\psi)(x) = \psi(-x) = \psi(x)$ for even functions.

$\rightarrow$ Compatible with $(O^B)^2 = 1$.

Recall:

We can interpret the Fourier transform as a change of basis:

$$\mathcal{F}(\psi)(p) = \langle p|\psi\rangle = \langle p| \int dx |x\rangle\langle x|\psi\rangle = \frac{1}{\sqrt{2\pi}} \int dx e^{-ipx}\psi(x)$$

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Fact:

Gauging can be expressed as

$$Z_{O^B(T_B)}[a] = \langle m, a|T_B\rangle = \langle m, a| \frac{1}{|H^1(\Sigma;\mathbb{Z}_2)|^{1/2}} \sum_b |e, b\rangle\langle e, b|T_B\rangle$$

$$= \frac{1}{|H^1(\Sigma;\mathbb{Z}_2)|^{1/2}} \sum_b (-1)^{\int a\cup b} Z_{T_B}[b]$$

where $\{|e, a\rangle\}$ and $\{|m, a\rangle\}$ are boundary conditions in $(2+1)$D TQFT, i.e. gauging in $(1+1)$ D can be described by $(2+1)$D TQFT.

SymTFT is a framework to understand symmetries in $(d+1)$D theories in terms of $(d+2)$D TQFTs.

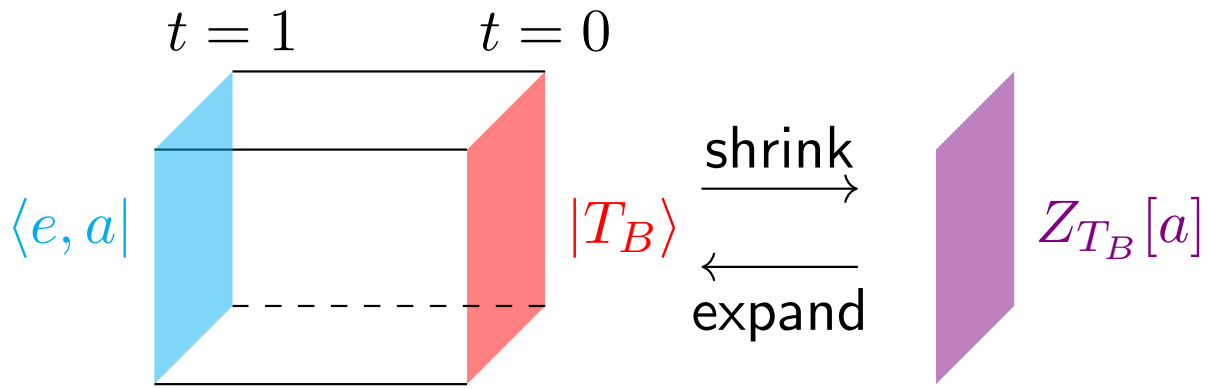


Figure: $|T_B\rangle/|e,a\rangle$ correspond to a physical/gapped boundary respectively, where the spacetime is given as $\Sigma \times [0,1]$.

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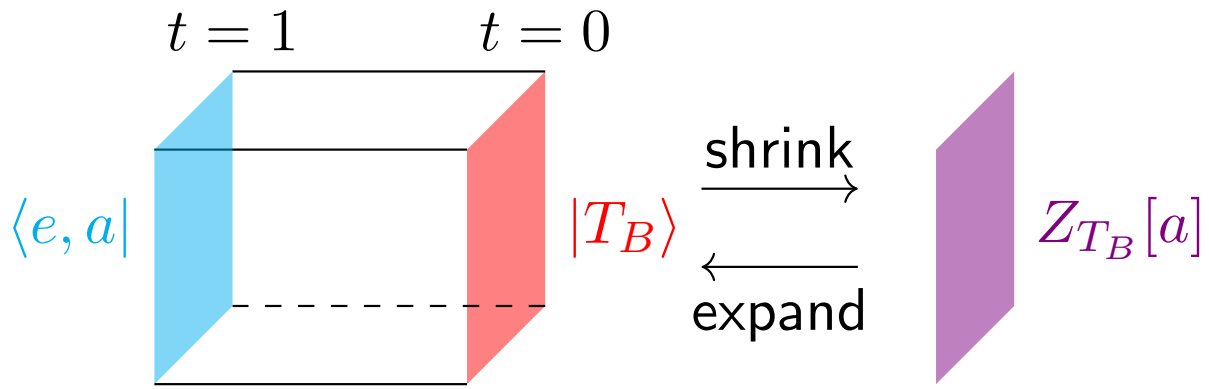


Figure: $|T_B\rangle/|e,a\rangle$ correspond to a physical/gapped boundary respectively, where the spacetime is given as $\Sigma \times [0,1]$.

Fact:
SymTFT for $\mathbb{Z}_2$ symmetry is a Toric code (i.e. $\mathbb{Z}_2$ gauge theory).

Toric code has line operators $\{1, E(a), M(a), F(a) := E(a)M(a)\}$ s.t.

$$E(a)E(b) = E(a+b), \quad M(a)M(b) = M(a+b).$$
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$\rightarrow$ Define the gapped boundary condition $|e, 0\rangle$ by condensing $E(a)$:

$$E(a)|e, 0\rangle = |e, 0\rangle \quad \text{for all } a \in H^1(\Sigma; \mathbb{Z}_2)$$

Then one can check that $|e, b\rangle := M(b)|e, 0\rangle$ is an eigenvector of $E(a)$:

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$\rightarrow$The Hilbert space on Σ assigned by $(2+1)$D TQFT is given as

$$\mathcal{H}(\Sigma) = \mathrm{span}_{\mathbb{C}}\{|e, a\rangle \mid a \in H^1(\Sigma; \mathbb{Z}_2)\}$$

We can define another boundary condition by condensing $M(a)$:

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$\rightarrow$ We construct SymTFT for $O^B, S_1^B, S_2^B, \mathrm{Fer}$ on non-orientable surfaces.

- Duality web on orientable surfaces
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We can study the duality from the perspectives of the Hilbert space!
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Here subscripts even/odd means the eigenvalues of $g : \mathcal{H}_{\text{all}} \to \mathcal{H}_{\text{all}}$:

$$g\psi = \begin{cases} \psi & \psi \in \mathcal{H}_{\text{even}} \oplus \mathcal{H}^g_{\text{even}} \\ -\psi & \psi \in \mathcal{H}_{\text{odd}} \oplus \mathcal{H}^g_{\text{odd}} \end{cases}$$

and superscripts mean untwisted/twisted by g.

T_B	untwisted : $\mathcal{H}$	twisted $\mathcal{H}^g$
even	S	U
odd	T	V

Table: Sectors of T_B on the circle S^1.

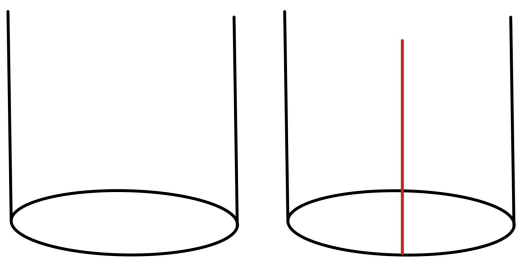


Figure: Untwisted/twisted Hilb sp.

Now we consider torus a partition function:

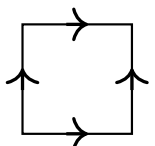
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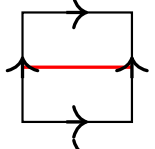
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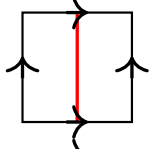
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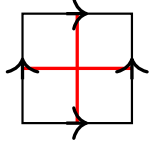
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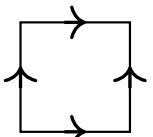
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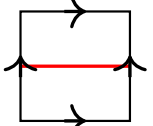
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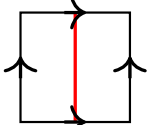
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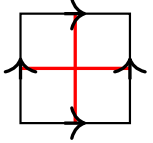
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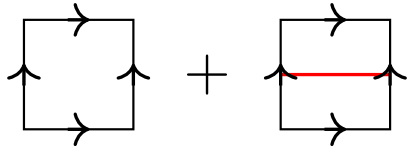
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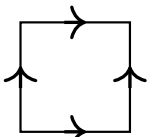
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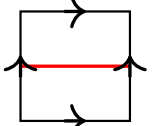
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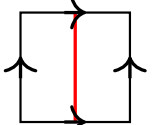
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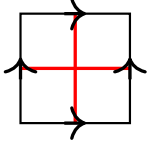
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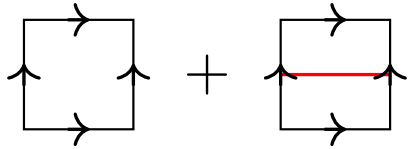
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$\rightarrow$ We can keep track of these sectors in the web of dualities!

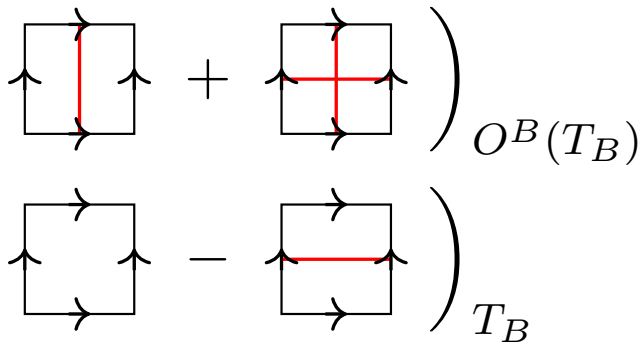
For example, we can calculate the even, twisted sector of $O^B(T_B)$ as

$$\mathrm{Tr}^{O^B(T_B)}_{\mathcal{H}^g}\left(\frac{1+g}{2}e^{-lH}\right)$$

$$= \frac{1}{2}\left(\quad + \quad \right)_{O^B(T_B)}$$

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which turns out to be the odd, untwisted sector of T_B T.

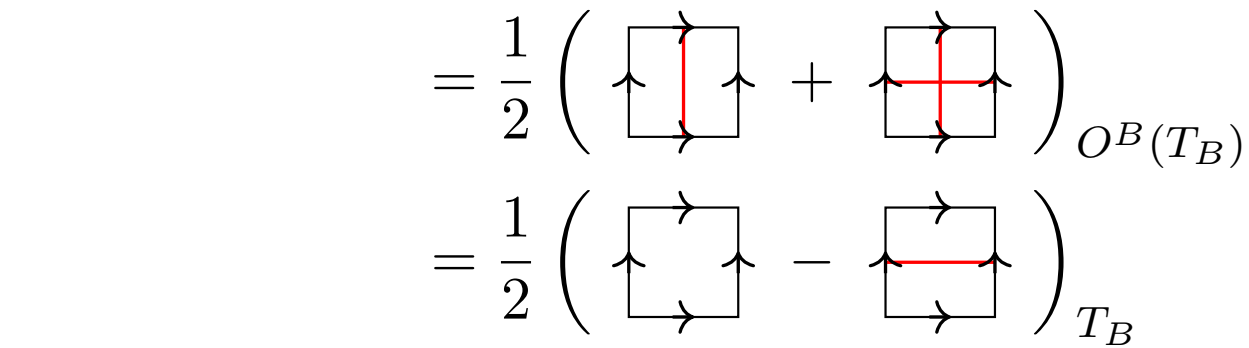
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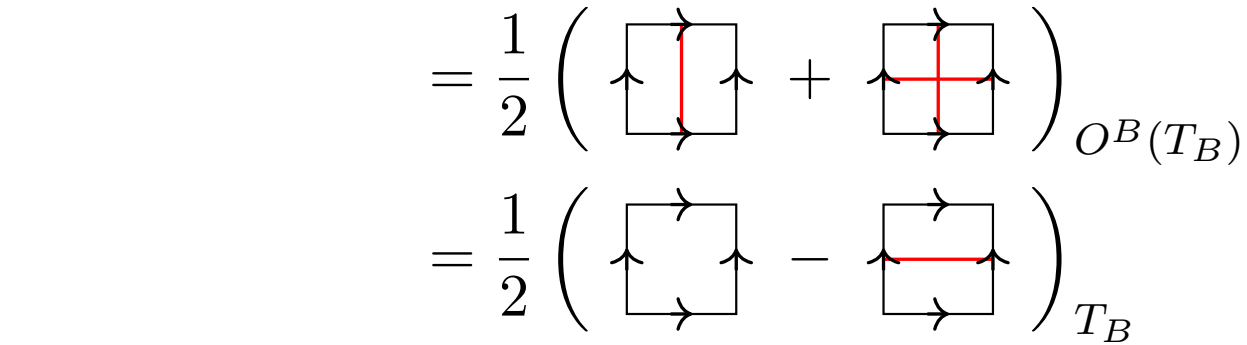
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$\rightarrow$ We can do the same for Fermionization.

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$$\mathcal{H}_{\text{all}} = \mathcal{H} \oplus \mathcal{H}^g \quad \text{with} \quad \begin{cases} \mathcal{H} = \mathcal{H}_{\text{even},+} \oplus \mathcal{H}_{\text{even},-} \oplus \mathcal{H}_{\text{odd},+} \oplus \mathcal{H}_{\text{odd},-} \\ \mathcal{H}^g = \mathcal{H}^g_{\text{even},+} \oplus \mathcal{H}^g_{\text{even},-} \oplus \mathcal{H}^g_{\text{odd},+} \oplus \mathcal{H}^g_{\text{odd},-} \end{cases}$$

Here subscripts $+/-$ mean the eigenvalues of $P : \mathcal{H}_{\text{all}} \to \mathcal{H}_{\text{all}}$:

$$P\psi = \begin{cases} \psi & \psi \in \mathcal{H}_+ \\ -\psi & \psi \in \mathcal{H}_- \end{cases}$$

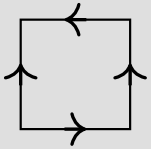
T_B		untwisted	twisted
even	$P = +1$	S_+	U_+
	$P = -1$	S_-	U_-
odd	$P = +1$	T_+	V_+
	$P = -1$	T_-	V_-

Table: Sectors of T_B on the circle S^1.

Suppose we have a torus partition function $Z_{T_B}[T^2;0] = \mathrm{Tr}_{\mathcal{H}}^{T_B}(e^{-lH})$

A partition function with an insertion of $P : \mathcal{H}_{\text{all}} \to \mathcal{H}_{\text{all}}$

$$Z_{T_B}[K;0] = \mathrm{Tr}_{\mathcal{H}}^{T_B}(Pe^{-lH}) =$$

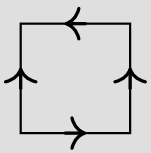


can be interpreted as a Klein bottle partition function.

Suppose we have a torus partition function $Z_{T_B}[T^2;0] = \mathrm{Tr}_{\mathcal{H}}^{T_B}(e^{-lH})$

A partition function with an insertion of $P : \mathcal{H}_{\text{all}} \to \mathcal{H}_{\text{all}}$

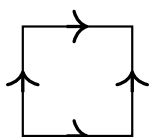
$$Z_{T_B}[K;0] = \mathrm{Tr}_{\mathcal{H}}^{T_B}(Pe^{-lH}) =$$



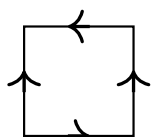
can be interpreted as a Klein bottle partition function.

In terms of $g : \mathcal{H}_{\text{all}} \to \mathcal{H}_{\text{all}}$ where $g \in \mathbb{Z}_2$, we obtain

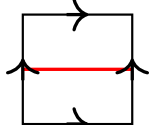
$$Z_{T_B}[T^2;0] = \mathrm{Tr}_{\mathcal{H}}^{T_B}(e^{-lH}) =$$



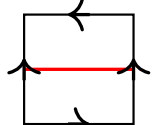
$$Z_{T_B}[K;0] = \mathrm{Tr}_{\mathcal{H}}^{T_B}(Pe^{-lH}) =$$



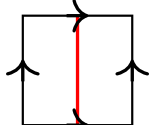
$$Z_{T_B}[T^2;x] = \mathrm{Tr}_{\mathcal{H}}^{T_B}(g\,e^{-lH}) =$$



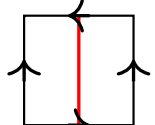
$$Z_{T_B}[K;x] = \mathrm{Tr}_{\mathcal{H}}^{T_B}(gPe^{-lH}) =$$



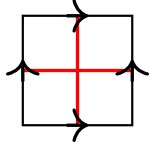
$$Z_{T_B}[T^2;y] = \mathrm{Tr}_{\mathcal{H}^g}^{T_B}(e^{-lH}) =$$



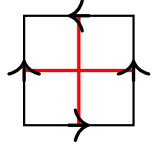
$$Z_{T_B}[K;y] = \mathrm{Tr}_{\mathcal{H}^g}^{T_B}(Pe^{-lH}) =$$



$$Z_{T_B}[T^2;x+y] = \mathrm{Tr}_{\mathcal{H}^g}^{T_B}(g\,e^{-lH}) =$$



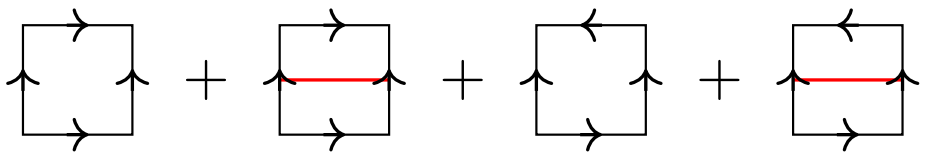
$$Z_{T_B}[K;x+y] = \mathrm{Tr}_{\mathcal{H}^g}^{T_B}(gP\,e^{-lH}) =$$



.

Then, the even, $+$, untwisted sector of T_B U_+ can be computed as

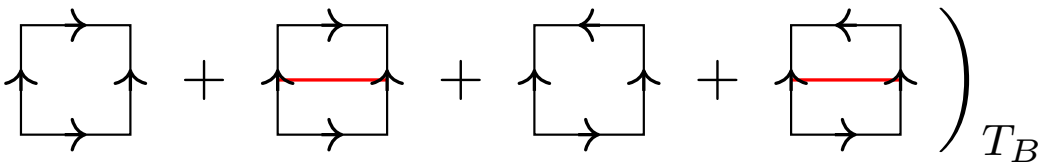
$$\mathrm{Tr}^{T_B}_{\mathcal{H}}\left(\frac{1+g}{2}\frac{1+P}{2}e^{-lH}\right) = \frac{1}{4}\Bigg($$



$$\Bigg)_{T_B}$$

Then, the even, $+$, untwisted sector of T_B U_+ can be computed as

$$\mathrm{Tr}^{T_B}_{\mathcal{H}}\left(\frac{1+g}{2}\frac{1+P}{2}e^{-lH}\right) = \frac{1}{4}\Big(\quad + \quad + \quad + \quad\Big)_{T_B}$$



$\rightarrow$ We can keep track of these sectors in the web of dualities!

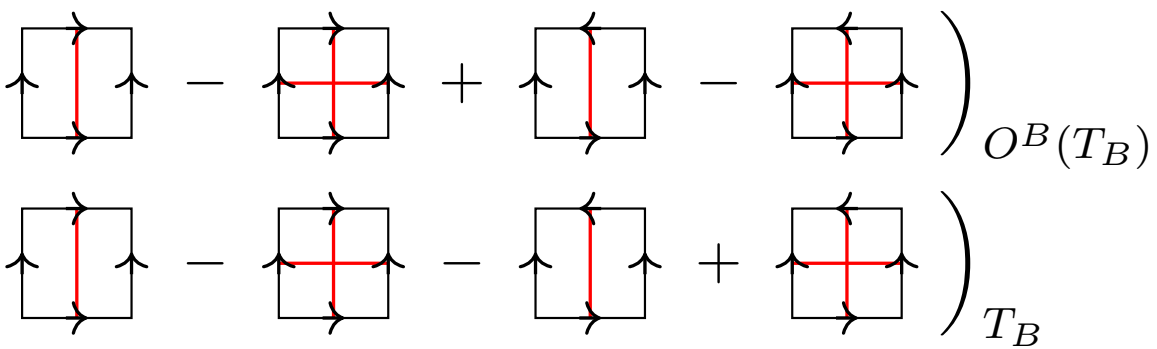
For example, we can calculate the odd, $+$, twisted sector of $O^B(T_B)$ as

$$\mathrm{Tr}^{O^B(T_B)}_{\mathcal{H}^g}\left(\frac{1-g}{2}\frac{1+P}{2}e^{-lH}\right)$$

$$= \frac{1}{4}\Big(\quad - \quad + \quad - \quad\Big)_{O^B(T_B)}$$

$$= \frac{1}{4}\Big(\quad - \quad - \quad + \quad\Big)_{T_B}$$

$$= \mathrm{Tr}^{T_B}_{\mathcal{H}^g}\left(\frac{1-g}{2}\frac{1-P}{2}e^{-lH}\right)$$



which turns out to be the odd, $-$, twisted sector of T_B V_-.

In this way, we give refined correspondence between T_B and $O^B(T_B)$ in terms of P eigenvalues:

T_B	$\mathcal{H}$	$\mathcal{H}^g$
even	S	U
odd	T	V

$\Downarrow$

T_B		$\mathcal{H}$	$\mathcal{H}^g$
even	$P=+1$	S_+	U_+
	$P=-1$	S_-	U_-
odd	$P=+1$	T_+	V_+
	$P=-1$	T_-	V_-

$O^B(T_B)$	$\mathcal{H}$	$\mathcal{H}^g$
even	S	T
odd	U	V

$\Downarrow$

$O^B(T_B)$		$\mathcal{H}$	$\mathcal{H}^g$
even	$P=+1$	S_+	T_+
	$P=-1$	S_-	T_-
odd	$P=+1$	U_+	V_-
	$P=-1$	U_-	V_+

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T_B	$\mathcal{H}$	$\mathcal{H}^g$
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odd	$P=+1$	T_+	V_+
	$P=-1$	T_-	V_-

$O^B(T_B)$	$\mathcal{H}$	$\mathcal{H}^g$
even	S	T
odd	U	V

$\Downarrow$

$O^B(T_B)$		$\mathcal{H}$	$\mathcal{H}^g$
even	$P=+1$	S_+	T_+
	$P=-1$	S_-	T_-
odd	$P=+1$	U_+	V_-
	$P=-1$	U_-	V_+

$\rightarrow$ How about fermionization?

In fermionic theories, we have $P^2 = (-1)^F$, i.e., $P^4 = 1$. Therefore, $\mathcal{H}_{\text{all}}$ can be decomposed in terms of $P = \pm 1, \pm i$, together with NS, R:

$$\mathcal{H}_{\text{all}} = \mathcal{H}_{\text{NS}} \oplus \mathcal{H}_{\text{R}} \quad \text{with} \quad \begin{cases} \mathcal{H}_{\text{NS}} = \mathcal{H}_{\text{NS},+1} \oplus \mathcal{H}_{\text{NS},+i} \oplus \mathcal{H}_{\text{NS},-1} \oplus \mathcal{H}_{\text{NS},-i}, \\ \mathcal{H}_{\text{R}} = \mathcal{H}_{\text{R},+1} \oplus \mathcal{H}_{\text{R},+i} \oplus \mathcal{H}_{\text{R},-1} \oplus \mathcal{H}_{\text{R},-i}. \end{cases}$$

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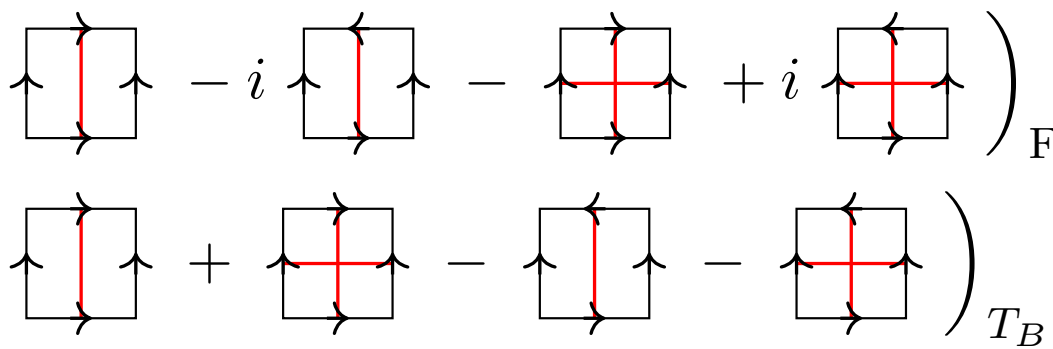
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For example, the R, $+i$ sector of $\mathrm{Fer}(T_B)$ can be computed as

$$\mathrm{Tr}^{\mathrm{Fer}(T_B)}_{\mathcal{H}_{\mathrm{R}}} \left(\frac{1 - iP - P^2 + iP^3}{4} e^{-lH} \right)$$

$$= \frac{1}{4} \Big(\quad - i \quad - \quad + i \quad \Big)_{\mathrm{Fer}(T_B)}$$

$$= \frac{1}{4} \Big(\quad + \quad - \quad - \quad \Big)_{T_B}$$



which turns out to be the even, $-$, twisted sector U_-.

With the same procedure, we obtain the following table:

T_B		$\mathcal{H}$	$\mathcal{H}^g$
even	$P=+1$	S_+	U_+
	$P=-1$	S_-	U_-
odd	$P=+1$	T_+	V_+
	$P=-1$	T_-	V_-

$O^B(T_B)$		$\mathcal{H}$	$\mathcal{H}^g$
even	$P=+1$	S_+	T_+
	$P=-1$	S_-	T_-
odd	$P=+1$	U_+	V_-
	$P=-1$	U_-	V_+

$S_1^B(T_B)$		$\mathcal{H}$	$\mathcal{H}^g$
even	$P=+1$	S_+	U_-
	$P=-1$	S_-	U_+
odd	$P=+1$	T_+	V_-
	$P=-1$	T_-	V_+

T_F	NS	R
$P=+1$	S_+	T_+
$P=-1$	S_-	T_-
$P=+i$	V_+	U_-
$P=-i$	V_-	U_+

$O^F(T_F)$	NS	R
$P=+1$	S_+	U_+
$P=-1$	S_-	U_-
$P=+i$	V_-	T_-
$P=-i$	V_+	T_+

$S_1^F(T_F)$	NS	R
$P=+1$	S_+	T_+
$P=-1$	S_-	T_-
$P=+i$	V_-	U_+
$P=-i$	V_+	U_-

Table: Summary of the dualities among sectors.

We refined the known dualities among S,T,U,V in terms of P!

- Duality web on orientable surfaces
  - $\mathbb{Z}_2$ symmetry and its gauging
  - $\mathbb{Z}_2$-quadratic refinement, Arf invariant, and spin structures
  - Fermionization and Bosonization
  - What changes on non-orientable surfaces
- Duality web on non-orientable surfaces
  - SymTFT
  - Sectors of Hilbert space
  - Group structure of topological manipulations

Recall:

We have topological manipulations on bosonic theories as follows:

$$Z_{O^B(T_B)}[a] = \frac{1}{|H^1(\Sigma;\mathbb{Z}_2)|^{1/2}} \sum_{b\in H^1(\Sigma;\mathbb{Z}_2)} (-1)^{\int a\cup b} Z_{T_B}[b]$$

$$Z_{S_1^B(T_B)}[a] = (-1)^{\int a\cup w_1} Z_{T_B}[a]$$

$$Z_{S_2^B(T_B)}[a] = (-1)^{\int w_1\cup w_1} Z_{T_B}[a]$$

Recall:

We have topological manipulations on bosonic theories as follows:

$$Z_{O^B(T_B)}[a] = \frac{1}{|H^1(\Sigma;\mathbb{Z}_2)|^{1/2}} \sum_{b\in H^1(\Sigma;\mathbb{Z}_2)} (-1)^{\int a\cup b} Z_{T_B}[b]$$

$$Z_{S_1^B(T_B)}[a] = (-1)^{\int a\cup w_1} Z_{T_B}[a]$$

$$Z_{S_2^B(T_B)}[a] = (-1)^{\int w_1\cup w_1} Z_{T_B}[a]$$

One can check that

$$(O^B)^2 = (S_1^B)^2 = (S_2^B)^2 = 1,\ (O^B S_1^B)^4 = S_2^B$$

Thus, the independent relations are as follows:

$$(O^B)^2 = (S_1^B)^2 = 1,\ (O^B S_1^B)^8 = 1$$

$\rightarrow$ The dihedral group D_8, i.e., $D_8 = \langle s, t \mid s^2 = t^2 = 1, (st)^8 = 1\rangle$

We have 16 elements in D_8 given as

$$(O^B S_1^B)^n,\ S_1^B (O^B S_1^B)^n \quad n = 0, 1, \cdots, 7$$

$\rightarrow$ We can easily see the D_8 structure on fermionic theories:

$$(O^F S_1^F)^n,\ S_1^F (O^F S_1^F)^n \quad n = 0, 1, \cdots, 7$$

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$\rightarrow$ We can easily see the D_8 structure on fermionic theories:

$$(O^F S_1^F)^n,\ S_1^F (O^F S_1^F)^n \quad n = 0, 1, \cdots, 7$$

As a result, we obtain the following web:

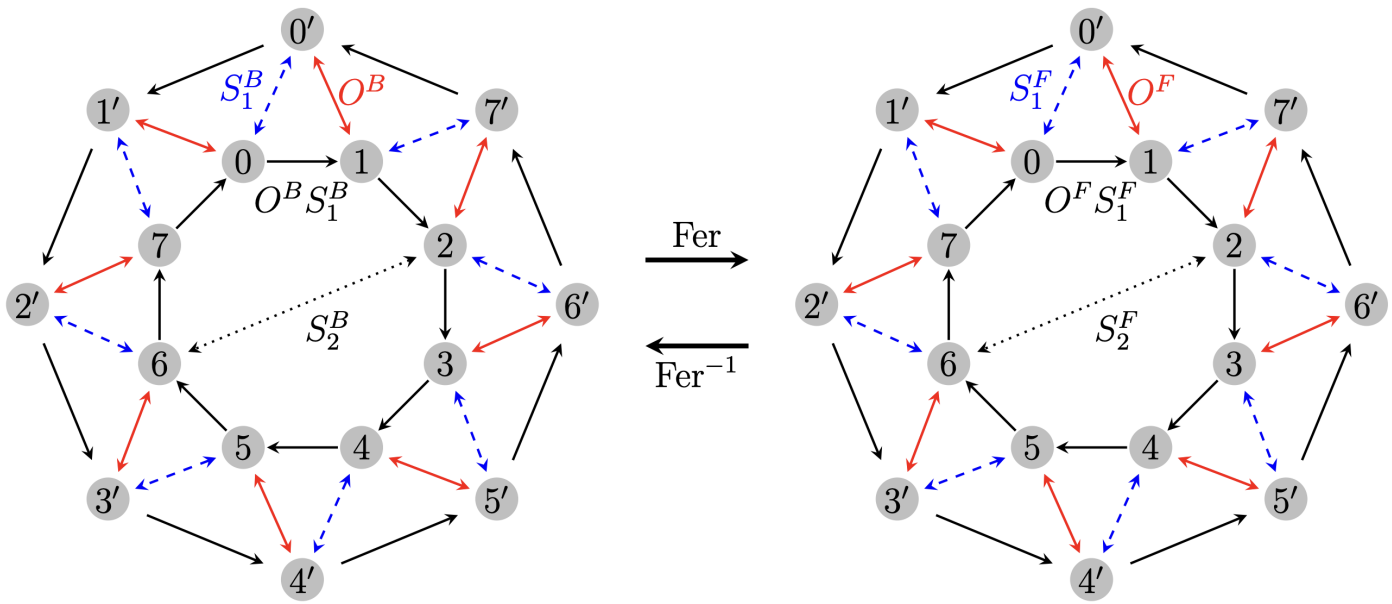
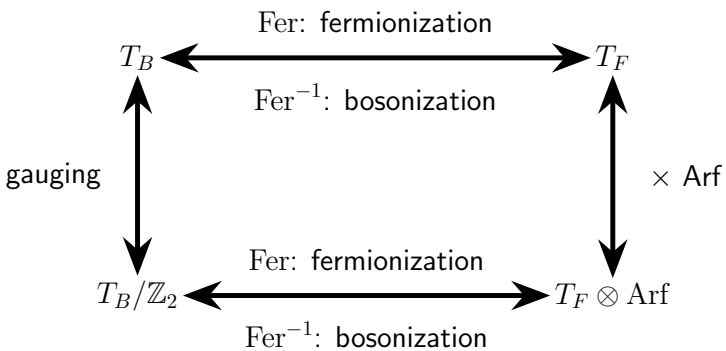


Figure: O^B : gauging $\mathbb{Z}_2$, S_1^B: stacking $(-1)^{\int a \cup w_1}$, S_2^B: stacking $(-1)^{\int w_1 \cup w_1}$

Results: Investigating non-orientable web of dualities in terms of

- SymTFT,
- Sectors of Hilbert space,
- Group structure of topological manipulations.



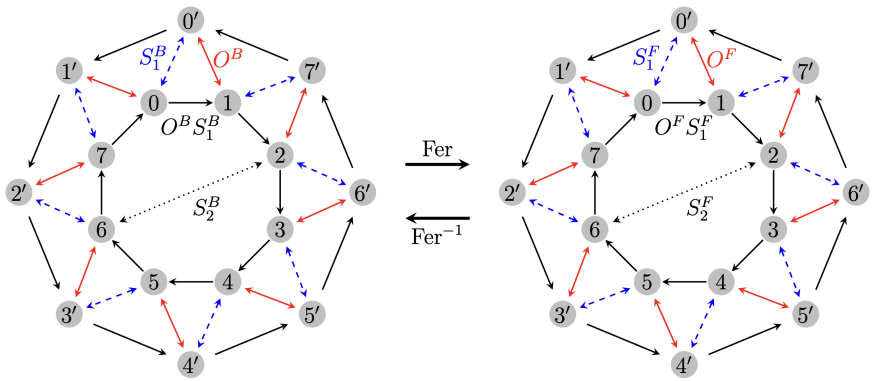


Figure: Orientable/non-orientable duality web

Future work:

- Extending to $\mathbb{Z}_n$ gauging and parafermionization.
- Studying the self duality of Ising/Majorana CFTs, etc...

Thank you!